

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to
Modernize the Electric Grid for a High
Distributed Energy Resources Future.

Rulemaking 21-06-017
(Filed June 24, 2021)

**RESPONSE OF THE VEHICLE-GRID INTEGRATION COUNCIL TO MOTION OF
ENPHASE ENERGY, INC. TO AMEND SCOPING RULING AND SCHEDULE TO
ADDRESS OPTIONAL FLEXIBLE CONNECTION AGREEMENTS AND THE
UNDERWRITERS' LABORATORY 3141 STANDARD FOR POWER CONTROL
SYSTEMS IN INVESTOR-OWNED UTILITIES' ENERGIZATION RULES**

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In accordance with Rule 11.1 of the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC")¹ hereby submits this response to the *Motion of Enphase Energy, Inc. to Amend Scoping Ruling and Schedule to Address Optional Flexible Connection Agreements ("FCA") and the Underwriters' Laboratory ("UL") 3141 Standard for Power Control Systems ("PCS") in Investor-owned Utilities' ("IOU") Energization Rules ("Motion")* filed by Enphase Energy on May 6, 2024.

I. INTRODUCTION.

It is widely acknowledged that meeting California's climate goals will require massive electrification of transportation, buildings, and industry. At the same time, California's electric

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., BorgWarner, Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, Customized Energy Solutions, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, FreeWire Technologies, Inc., GridWiz, Kaluza, Landis + Gyr, Leapfrog Power, Inc., Nuvve Holding Corporation, Peak Power, PowerFlex, Qcells, Stellantis N.V., Sumitomo Electric, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Hoosier Energy, Peninsula Clean Energy, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters.
(<https://www.vgicouncil.org/>)

grid has not yet been built out to accommodate this extra load. For this reason, the Commission has turned its focus to improving planning and bringing additional generation, transmission, and distribution resources online. However, bringing all of these resources online will take significant amounts of time, and it is becoming increasingly clear that interim measures are needed to allow for electrification ahead of grid upgrades. Additionally, the costs of these upgrades will impact ratepayers and additional cost-saving measures of electrification will need to be pursued to ensure that electricity remains affordable.

Enphase's Motion highlights the need to leverage tools that are currently available to allow for faster and cheaper electric energization. VGIC supports Enphase's Motion and agrees that Flexible Connection Agreements ("FCA") enabled by power control systems ("PCS") should be leveraged by the Commission and is appropriately considered in Track 3 of this High Distributed Energy Resource ("DER") proceeding. VGIC offers additional comments on our support below.

II. VGIC SUPPORTS ENPHASE'S MOTION TO INCLUDE POWER CONTROL SYSTEMS, UL 3141, AND FLEXIBLE CONNECTION AGREEMENTS IN TRACK 3 OF THIS PROCEEDING.

In recognition of the goals and challenges highlighted above, the Smart Inverter Operationalization Working Group ("SIOWG") in this proceeding determined that, "operational flexibility [should be] the highest priority, namely the ability of the DSOs in the high DER future to flexibly optimize the use of existing capacity, allowing more rapid connections of DER and loads, while still maintaining grid safety and reliability."² VGIC agrees that this is the most important smart inverter operationalization use case, particularly for the development of Flexible Connection Agreements or *Limited Load Profiles* ("LLP"). While the report discussed leveraging operational flexibility in "the high DER future", *many of these techniques and technologies can*

² SIOWG Report at i.

and should be leveraged today or in the very near future to enable customers to energize load more quickly and reduce ratepayer costs. As discussed by Enphase, the PCS that would create operational flexibility is already widely used to control DERs and is a technology understood by utilities.³ The UL standard that would allow PCS to be used for LLPs is expected to be published next month, June 2024, and the standards development process relies heavily on utility input.⁴

With this in mind, it appears the current barrier to deployment is largely regulatory, since the development of LLPs may require updates to utility rules and tariffs. Rules 2, 15, 16, 29, and 45 may need some modification and specific tariff language and terms and conditions need to be developed to enable an LLP, just as was adopted recently for Limited Generation Profiles (“LGP”). VGIC supports Enphase’s proposed question to be added to the Track 3, Phase 1 scope to address these issues, “Should the investor-owned utilities be ordered to amend their energization rules to allow customers to elect optional flexible connection agreements that use UL 3141-certified power control systems to implement a Limited Load Profile that avoids distribution upgrades, thus enabling rapid and cost-effective project deployment and operations?”⁵ We also agree that tariff provisions and technical requirements for LLPs should be developed.

Development of these updated rules and tariffs will take time and conversations should begin as soon as possible with workshops or other discussion since it requires significant utility and stakeholder engagement to develop any necessary tariffs and programs. For example, the SIOGW had meetings for two years to develop these report recommendations and the development of the Limited Generation Profiles, which creates a flexible export agreement, took 3.5 years to finalize. Therefore, while the updated UL 3141 standard has not been published yet, workshops

³ Enphase Motion at 4.

⁴ Enphase Motion at 6.

⁵ Enphase Motion at 7.

and other stakeholder discussions must begin now to inform the forthcoming Staff Proposal that will cover the SIOGW report and these issues of operational flexibility. Many regulatory details can be worked through while the standard is finalized.

III. DEVELOPMENT OF FLEXIBLE CONNECTION AGREEMENTS SHOULD BE CLOSELY COORDINATED WITH THE ENERGIZATION TIMELINE RULEMAKING.

The Commission is currently looking to address difficulties connecting new load to the grid by setting target maximum and average energization timelines in Rulemaking (“R.”) 24-01-018. As VGIC expressed in comments on Phase 1 of that Rulemaking, the critical topic of flexible service connection, which includes elements like FCAs and LLPs, will help the utilities achieve reasonable timelines for customer energization.⁶ In the Energization Timelines OIR, VGIC recommends that the utilities include reporting about whether customers were offered and/or will use flexible service connections. Once FCAs and LLPs are developed, recording which customers leverage these agreements should be included in the utility energization timelines reporting.

Additionally, reporting which factors are creating energization limitations, what is lengthening the energization timeline, and data on which flexible service connection types and technologies are most common can help to inform the ongoing development of FCAs and LLPs. Therefore, VGIC recommends close coordination between both this High DER proceeding and the Energization Timeline proceeding to ensure that the Commission can gather data on how flexible service connections are used and how to leverage FCAs to create faster energization timelines for customers.

IV. CONCLUSION.

⁶ VGIC Comments at 6.

VGIC appreciates the opportunity to provide this response to the motion. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

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VEHICLE-GRID INTEGRATION COUNCIL

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